Online Appendix for: On Existence and Uniqueness of Equilibrium in a Class of Noisy Rational Expectations Models

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1 The importance of CARA utility and exponential family beliefs

The combination of Assumption 1 on CARA utility and Assumption 3 on exponential family beliefs are important for both the equilibrium characterization and the uniqueness result because they allow one to identify a sufficient statistic for the equilibrium price without first identifying the price function. As noted in the text, the assumptions guarantee that the informed investor's trading aggressiveness $\frac{\partial X_I}{\partial s_I}(s_I, p)$ is independent of the price. Since the information revealed by the price depends on this aggressiveness but the aggressiveness does not depend on the price, it follows that the information content of the price can be pinned down independently. With this information content pinned down, there is no channel by which non-uniqueness can occur, at least among smooth price functions. On the other hand, if the informed trading aggressiveness depended on the price, it would open up the possibility of multiple self-fulfilling equilibria across which the price is more or less informative.

It turns out that the combination of CARA utility and exponential family beliefs is necessary and sufficient for additively separable demand functions. Regardless of the conditional distribution of payoffs, any risk-averse utility function outside of the CARA class will exhibit wealth effects and therefore trading aggressiveness will in general depend on the price due to the effect of price on wealth. I establish this fact in the following Proposition in which I consider the portfolio problem of an investor with twice continuously differentiable utility function $u(\cdot)$ defined on some set $U \subseteq \mathbb{R}$ and which is increasing and concave on that domain. The investor is endowed with y_0 units of the risk-free asset and x_0 shares of the risky asset. I consider the endowment explicitly because without CARA utility, asset demand may depend on the initial wealth. Further, I assume that (a) it is permissible to pass derivatives through expectations and (b) the first-order condition is necessary for an optimum. Sufficient conditions guaranteeing these assumptions are straightforward to determine, e.g., (a) asset payoffs with tails that die off sufficiently quickly and (b) the set of feasible portfolios being an open set. As an enumeration of such conditions on primitives are not important for the result, I omit them.

Proposition 1.1. *Suppose that the utility function $u : U \to \mathbb{R}$ is twice continuously differentiable.*

If, for an arbitrary initial endowment (x_0, y_0), the informed investor's demand function can be written in the form

$$X_I(s_I, p, y_0, x_0) = a(s_I, y_0, x_0) - b(p, y_0, x_0)$$

for some functions a and b, then the investor's utility function must exhibit constant absolute risk aversion on U.

Proof. Suppose that the demand function takes the given additively separable form. I begin by deducing some properties that must be satisfied by the functions a and b. By assumption, the demand function must satisfy the first-order condition

$$\int_{\mathcal{V}} (v-p)u'(y_0 + x_0 p + X_I(\cdot)(v-p)) f_{\widetilde{V}|\widetilde{S}_I}(v|s_I)\, dv = 0,$$

and under the differentiability assumption on the integral and smoothness assumption on the utility function, the implicit function theorem implies that the demand function will also be continuously differentiable in the price and initial endowment.

Notice also that the functions $a(\cdot)$ and $b(\cdot)$ which define $X_I(\cdot)$ must depend on y_0 and x_0 only through the initial wealth $y_0 + x_0 p$. To see this, notice that for fixed (s, p), the FOC holds for any other initial endowment (x_0', y_0') satisfying $y_0' + x_0' p = y_0 + x_0 p$. Hence, the functions a and b each depend on only two arguments, one of which is the initial wealth, $a(s_I, y_0 + x_0 p)$ and $b(p, y_0 + x_0 p)$. Additionally, since the demand function is assumed to be additively separable in s_I and p, it must be the case that $a(s_I, y_0 + x_0 p)$ is itself is additively separable in s and the initial wealth. Hence, without loss of generality, any terms in a that depend only on the initial wealth can absorbed into the function b and the demand function can written in the form $X_I(s_I, p, y_0, x_0) = a(s_I) - b(p, y_0 + x_0 p)$ for some functions a and b, where b is continuously differentiable in both of its arguments. With this demand function in hand, I now turn to the proof of the main result.

Under the differentiability and smoothness assumptions, the implicit function theorem guarantees the existence of $\frac{d}{dp} X_I(\cdot) = -b_p(p, y_0 + x_0 p) - x_0 b_{w_0}(p, y_0 + x_0 p)$, where I use subscripts p and w_0 to denote partial derivatives with respect to the first and second arguments, respectively. Differentiating the FOC totally with respect to p and rearranging gives

$$\begin{aligned}
&- x_0 b_{w_0}(p, y_0 + x_0 p) - b_p(p, y_0 + x_0 p) \\
&= \frac{\int_{\mathcal{V}} u'(y_0 + x_0 p + X_I(\cdot)(v-p)) f_{\widetilde{V}|\widetilde{S}_I}(v|s_I)\, dv}{\int_{\mathcal{V}} (v-p)^2 u''(y_0 + x_0 p + X_I(\cdot)(v-p)) f_{\widetilde{V}|\widetilde{S}_I}(v|s_I)\, dv} \\
&\quad - \frac{\int_{\mathcal{V}} (x_0 - X_I(\cdot))(v-p) u''(y_0 + x_0 p + X_I(\cdot)(v-p)) f_{\widetilde{V}|\widetilde{S}_I}(v|s_I)\, dv}{\int_{\mathcal{V}} (v-p)^2 u''(y_0 + x_0 p + X_I(\cdot)(v-p)) f_{\widetilde{V}|\widetilde{S}_I}(v|s_I)\, dv}
\end{aligned}$$

Let $E(s_I) \equiv \mathbb{E}[\widetilde{V}|\widetilde{S}_I = s_I]$ and evaluate this expression at $p = E(s_I)$. At this point, the optimal

demand is zero since the investor perceives the asset as having a zero risk premium. Hence,

$$
\begin{aligned}
&- x_0 b_{w_0}(E(s_I), y_0 + x_0 E(s_I)) - b_p(E(s_I), y_0 + x_0 E(s_I)) \\
&= \frac{u'(y_0 + x_0 E(s_I)) \int_{\mathcal{V}} f_{\widetilde{V}|\widetilde{S}_I}(v|s_I)\, dt - x_0 u''(y_0 + x_0 E(s_I)) \int_{\mathcal{V}} (v - E(s_I)) f_{\widetilde{V}|\widetilde{S}_I}(v|s_I)\, dv}{u''(y_0 + x_0 E(s_I)) \int_{\mathcal{V}} (v - E(s_I))^2 f_{\widetilde{V}|\widetilde{S}_I}(v|s_I)\, dv} \\
&= \frac{-1}{\frac{-u''(y_0+x_0E(s_I))}{u'(y_0+x_0E(s_I))} \operatorname{Var}(\widetilde{V}|s_I)}, \qquad (1)
\end{aligned}
$$

where the second equality uses $\int_{\mathcal{V}} f_{\widetilde{V}|\widetilde{S}_I}(v|s_I)\, dt = 1$, $\int_{\mathcal{V}} (v - E(s_I)) f_{\widetilde{V}|\widetilde{S}_I}(v|s_I)\, dv = 0$, and $\int_{\mathcal{V}} (v - E(s_I))^2 f_{\widetilde{V}|\widetilde{S}_I}(v|s_I)\, dv = \operatorname{Var}(\widetilde{V}|s_I)$.

The intuition behind this equation is that near points where the investor finds it optimal to hold zero risky assets, she is locally a mean-variance optimizer, choosing a portfolio $X_I \approx \frac{E[\widetilde{V}|\widetilde{S}_I = s_I] - p}{\frac{-u''(y_0+x_0E(s_I))}{u'(y_0+x_0E(s_I))} \operatorname{Var}(\widetilde{V}|s_I)}$. Let $\tau(y_0 + x_0 E(s_I)) \equiv \frac{-u'(y_0+x_0E(s_I))}{u''(y_0+x_0E(s_I))}$ be the coefficient of absolute risk tolerance evaluated at $y_0 + x_0 E(s_I)$.

Rearranging eq. (1) implies that for any arbitrary y_0, x_0, s_I, the risk tolerance τ must satisfy

$$
\begin{aligned}
&\tau(y_0 + x_0 E(s_I)) \\
&= \operatorname{Var}(\widetilde{V}|s_I) \left[x_0 b_{w_0}(E(s_I), y_0 + x_0 E(s_I)) + b_p(E(s_I), y_0 + x_0 E(s_I)) \right].
\end{aligned}
$$

Let $w_0 = y_0 + x_0 E(s_I)$ denote the initial wealth at this price (note that for any fixed $s_I \in \mathcal{S}$, w_0 can be chosen to be an arbitrary point in the domain U by appropriate choice of $(y_0, x_0) \in \mathbb{R}^2$) and simplify the previous expression

$$
\tau(w_0) = \operatorname{Var}(\widetilde{V}|s_I) \left[x_0 b_{w_0}(E(s_I), w_0) + b_p(E(s_I), w_0) \right].
$$

Since the left-hand side of this expression depends only on w_0, the equality must still hold when (y_0, x_0) is replaced by any $(y_0', x_0') \in \mathbb{R}^2$ such that $y_0' + x_0' E(s_I) = w_0$. Given that $\operatorname{Var}(\widetilde{V}|s_I) > 0$, the only way this can be true is if $b_{w_0}(E(s_I), w_0) = 0$. Otherwise the value of the right-hand side would vary with the initial endowment of shares, x_0. Since s_I and w_0 were chosen arbitrarily, it follows that $b_{w_0} = 0$ everywhere and therefore b is constant with respect to its second argument. With that fact, we can suppress the second argument in b and the most recent displayed expression becomes

$$
\tau(w_0) = \operatorname{Var}(\widetilde{V}|s_I) b_p(E(s_I)).
$$

The only way that this expression can hold for arbitrary $w_0 \in U$ and $s_I \in \mathcal{S}$ is if τ is constant with respect to w_0 and if the right-hand side is constant with respect to s_I. Since τ is the absolute risk

tolerance, it follows that the agent's utility function must be CARA.[1] □

I now show that the exponential family assumption is necessary and sufficient for a CARA investor to have an additively separable demand function. Sufficiency was shown in Lemma 3.1, which proved that under Assumptions 1 – 4 informed investor demand functions are additively separable in s_I and p

$$X_I(s_I, p) = \tau_I(k_I s_I - G_I(p)).$$

Recall from the discussion following Assumption 3 that the fact that the demand function is linear in s_I is without loss of generality as one can always transform the signal to achieve this. Thus, this result implicitly captures more general forms of additive separability.

I record the converse this in the following Proposition. The result relies on a characterization of exponential families in terms of the functional form of the moment generating function that appears in the theoretical statistics literature (Hoffmann and Schmidt, 1982; Sampson, 1975).

Proposition 1.2. *Suppose that Assumptions 1 and 2 hold. Suppose that for all $s_I \in \mathcal{S}$ the conditional mgf of $\widetilde{V}$ converges only on an open interval.*

If the informed investor's demand function is of the form

$$X_I(s_I, p) = \tau_I(a s_I - G_I(p)),$$

where $a > 0$ and G_I is some strictly increasing function mapping $(\underline{V}, \overline{V})$ to $\mathbb{R}$, then the conditional distribution of $\widetilde{V}$ given $\widetilde{S}_I = s_I$ is of the exponential family form of eq. (1) with $k_I = a$, $g_I' = G_I^{-1}$, and the support $\mathcal{G}_I$ of g_I is an open interval satisfying $k_I \mathcal{S} \subseteq \mathcal{G}_I$.

Proof of Proposition 1.2. Under the assumption that the conditional mgf converges only on an open set, Lemma A.4 implies that the first-order condition is necessary and sufficient for an optimum. The first-order condition that defines the informed investor's demand function X_I is

$$\mathbb{E}\left[(\widetilde{V} - p)\exp\left\{-\frac{1}{\tau_I}X_I(s_I, p)(\widetilde{V} - p)\right\}|\widetilde{S}_I = s_I\right] = 0.$$

[1] To see that this condition is consistent with the CARA-exponential family setting, notice that in that case, Lemma A.3 gives

$$E(s_I) = g_I'(k_I s_I)$$
$$\mathrm{Var}(\widetilde{V}|\widetilde{S}_I = s_I) = g_I''(k_I s_I),$$

and one has

$$b_p(E(s_I)) = \frac{\partial}{\partial p}\tau_I G_I(p)\bigg|_{p=E(s_I)} = \tau_I \frac{\partial}{\partial p}(g_I')^{-1}(p)\bigg|_{p=E(s_I)} = \tau_I \frac{1}{g_I''((g_I')^{-1}(E(s_I)))} = \tau_I \frac{1}{g''(k_I s_I)}.$$

Hence $\mathrm{Var}(\widetilde{V}|s_I)b_p(E(s_I)) = \tau_I$.

Recalling that the conditional moment generating function of $\widetilde{V}$ is

$$M_{\widetilde{V}|\widetilde{S}_I}(u|s_I) = \mathbb{E}\left[\exp\{u\widetilde{V}\}|\widetilde{S}_I = s_I\right],$$

the FOC can be rewritten in terms of conditional mgfs as

$$\frac{M'_{\widetilde{V}|\widetilde{S}_I}(-\frac{1}{\tau_I}X_I(s_I,p)|s_I)}{M_{\widetilde{V}|\widetilde{S}_I}(-\frac{1}{\tau_I}X_I(s_I,p)|s_I)} = p.$$

To save space, let $c_{\widetilde{V}|\widetilde{S}_I}(u|s_I) = \log M_{\widetilde{V}|\widetilde{S}_I}(u|s_I)$ denote the so-called conditional cumulant generating function and write the FOC compactly as

$$c'_{\widetilde{V}|\widetilde{S}_I}\left(-\frac{1}{\tau_I}X_I(s_I,p)|s_I\right) = p.$$

By inspection, if the demand function is of the given additively separable form, $X_I(s_I,p) = \tau_I(as_I - G_I(p))$, then the function $c'_{\widetilde{V}|\widetilde{S}_I}(u|s_I)$ must satisfy

$$c'_{\widetilde{V}|\widetilde{S}_I}(u|s_I) = G_I^{-1}(u + as_I)$$

for all $s_I \in \mathcal{S}$ and $u + as_I$ in the (open) set on which the mgf converges. By assumption, G_I is strictly increasing and defined on an interval, and an identical argument to that used for the uninformed investor in Lemma A.7 implies that the demand function must be continuous in p so that G_I is also continuous. Hence, the inverse that appears in the most recent expression is also strictly increasing, defined on an interval, and continuous.

By integrating with respect to u it is equivalent to show that $c_{\widetilde{V}|\widetilde{S}_I}$ itself satisfies

$$c_{\widetilde{V}|\widetilde{S}_I}(u|s_I) = \hat{G}_1(u + as_I) + \hat{G}_2(as_I)$$

where $\hat{G}_1 = \int G_I^{-1}$ is the antiderivative of G_I^{-1} and $\hat{G}_2$ is some function of as_I. Since the cumulant generating function satisfies $c_{\widetilde{V}|\widetilde{S}_I}(0|s_I) = \log M_{\widetilde{V}|\widetilde{S}_I}(0|s_I) = \log 1 = 0$ for all s_I, one can conclude that the function $\hat{G}_2$ must be given by $\hat{G}_1(as_I)$, leading to the relation

$$c_{\widetilde{V}|\widetilde{S}_I}(u|s_I) = \hat{G}_1(u + as_I) - \hat{G}_1(as_I),$$

or equivalently

$$M_{\widetilde{V}|\widetilde{S}_I}(u|s_I) = \frac{\exp\{\hat{G}_1(u + as_I)\}}{\exp\{\hat{G}_1(as_I)\}}. \tag{2}$$

Thus, a linear demand function implies that the conditional mgf must take this particular form.

Define $k_I = a$ and let $\mathcal{G}_I$ denote the support of the function $\hat{G}_I$. I will now show that $k_I\mathcal{S} \subseteq \mathcal{G}_I$ and $\mathcal{G}_I$ is an open interval. Since it was assumed that the conditional mgf exists for any $s_I \in \mathcal{S}$, the expression for the conditional mgf in eq. (2) implies that $k_I s_I \in \mathcal{G}_I$ since otherwise $\hat{G}_1(k_I s_I)$ would not exist. To show that $\mathcal{G}_I$ is an interval, take any $s_I \in \mathcal{S}$ and $\omega_1, \omega_2 \in \mathcal{G}_I$. Without loss of generality suppose $\omega_1 < \omega_2$. It suffices to show that any $\omega_3 \in [\omega_1, \omega_2]$ lies in $\mathcal{G}_I$. Take $\omega_3 \in [\omega_1, \omega_2]$, and for $j \in \{1, 2, 3\}$ let $u_j = \omega_j - k_I s_I$. It follows that $u_1 < u_3 < u_2$. Both

$$M_{\widetilde{V}|\widetilde{S}_I}(u_1|s_I) = \frac{\exp\{\hat{G}_1\left(u_1 + k_I s_I\right)\}}{\exp\{\hat{G}_1(k_I s_I)\}} = \frac{\exp\{\hat{G}_1(\omega_1)\}}{\exp\{\hat{G}_1(k_I s_I)\}} \quad \text{and}$$
$$M_{\widetilde{V}|\widetilde{S}_I}(u_2|s_I) = \frac{\exp\{\hat{G}_1\left(u_2 + k_I s_I\right)\}}{\exp\{\hat{G}_1(k_I s_I)\}} = \frac{\exp\{\hat{G}_1(\omega_2)\}}{\exp\{\hat{G}_1(k_I s_I)\}}$$

are finite since $\omega_1, \omega_2 \in \mathcal{G}_I$. Since mgfs are defined on intervals whenever they exist, and $u_3 \in [u_1, u_2]$, it follows that

$$M_{\widetilde{V}|\widetilde{S}_I}(u_3|s_I) = \frac{\exp\{\hat{G}_1(u_3 + k_I s_I)\}}{\exp\{\hat{G}_1(k_I s_I)\}} = \frac{\exp\{\hat{G}_1(\omega_3)\}}{\exp\{\hat{G}_1(k_I s_I)\}}$$

is finite and therefore that ω_3 must lie in $\mathcal{G}_I$. Finally, to prove that $\mathcal{G}_I$ is open, take any s_I and consider the set of $u \in \mathbb{R}$ on which the conditional mgf is finite:

$$\{u \in \mathbb{R} : M_{\widetilde{V}|\widetilde{S}_I}(u|s_I) < \infty\} = \{u \in \mathbb{R} : u + k_I s_I \in \mathcal{G}_I\}.$$

Under stated assumption that for each $s_I \in \mathcal{S}$ the conditional mgf converges only on an open set, it follows that $\mathcal{G}_I$ must itself be open.

It remains to prove that the conditional distribution of $\widetilde{V}$ takes the exponential family form. Note that because $k_I\mathcal{S} \subseteq \mathcal{G}_I$ and $\mathcal{G}_I$ is an interval, it follows that the convex hull of $k_I\mathcal{S}$ is contained in $\mathcal{G}_I$. Hence, $\hat{G}_1$ is well-defined on $\text{conv}(k_I\mathcal{S})$. Since the conditional mgf is of the form in eq. (2) and the function $\hat{G}_1$ is defined on $\text{conv}(k_I\mathcal{S})$, Theorem 1 of Hoffmann and Schmidt (1982) implies that the conditional distribution must be of the exponential family form in eq. (1) with $g_I(k_I s_I) = \hat{G}_1\left(k_I s_I\right)$, which implies $g_I' = G_I^{-1}(k_I s_I)$. As the proof of this result is technical and contains no economic insight, I do not repeat it here and refer the interested reader to the original paper.[2] □

2 Existence and uniqueness in a limiting case of Example 2

In this Section, I provide existence and uniqueness proofs for a limiting case of Example 2 with additive Gaussian signals in which the number of informed investors is large. More specifically,

[2] See also Theorem 2 of Sampson (1975) who proves the result under slightly stronger assumptions.

following Admati (1985) I analyze directly an economy with a continuum of investors.[3] In addition to assuring uniqueness among continuous equilibria, assuming an infinite number of investors also simplifies the expression for the equilibrium price function. As a continuum assumption is typical in applied work focusing on information aggregation, the results in this Section are more directly applicable to much of the literature than are the finite-investor results from Section 6.

As in the finite-investor case of Example 2 the payoff follows an arbitrary distribution $F_{\widetilde{V}}(v)$. There is a continuum of informed investors indexed by i, with risk tolerances $\tau_i > 0$ which are bounded from above. Each informed investor observes an additive signal of the form

$$\widetilde{S}_i = \widetilde{V} + \widetilde{\varepsilon}_i,$$

where $\widetilde{\varepsilon}_i \sim N(0, \sigma_i^2)$ are independently distributed. For simplicity, there are no uninformed investors, though they can be easily accommodated by allowing a positive measure of the σ_i^2 to tend to ∞. The supply shock is distributed $N(0, \sigma_Z^2)$.[4] Following Admati (1985), I assume that an exact law of large numbers holds so that when signals are aggregated across investors, individual errors wash out. That is, supposing that the σ_i^2 are bounded and c_i is a bounded set of constants, the following holds

$$\int_i c_i \widetilde{S}_i \, di \equiv \widetilde{V} \int_i c_i \, di.$$

There are well-known technical issues involved in dealing with a continuum of random variables (Feldman and Gilles, 1985; Judd, 1985) that render this a delicate matter to treat rigorously. Thus, as is typical in the noisy rational expectations literature, I take law of large numbers as a convention. See Admati (1985) and Vives (2008) for further details and a limiting argument that provides one justification for the assumption.

In this setting, an equilibrium is of the generalized linear form if it can be written as $P(L(v, z))$ for some linear function $L : \mathcal{V} \times \mathcal{Z} \to \mathbb{R}$ and some monotone function $P : L(\mathcal{V} \times \mathcal{Z}) \to \mathbb{R}$. This is the natural analogue of Definition 3 in a large economy in which idiosyncratic errors do not affect the price. Therefore, after normalizing the coefficient on z to -1, solving for the function L reduces to finding a single coefficient, namely that of v.

[3] In the case of jointly normally distributed random variables, it is well-known that the price function in the continuum economy is identical to that in the "large economy" limit. See Hellwig (1980) for details of the limiting procedure.

[4] This assumption can be substantially generalized to permit (at least) any distribution that is logconcave, as required in Assumption 7 to guarantee a monotone price function. However, doing so complicates the analysis since the resulting conditional distribution is not precisely in the exponential family form from Assumption 10 and produces demand functions that are only additively separable in s_i and some potentially nonlinear function of ℓ and the price p. This poses no problem for construction of equilibrium in the continuum case, unlike in the finite investor case, but would require the introduction of additional notation to capture the more complicated form of the demand functions.

The following characterization Proposition is analogous to Proposition 6.2 in the finite investor case.

Proposition 2.1. *Suppose that Assumption 1 holds in the continuum economy with an additive signal structure with normally distributed errors. Let* $L^*(v,z) = \left(\int_i \frac{\tau_i}{\sigma_i^2}\,di\right) v - z$, *and let* G *be the aggregate (risk-tolerance-weighted) price reaction function corresponding to* $a^* = \int_i \frac{\tau_i}{\sigma_i^2}\,di$. *That is,*

$$G(p) \equiv \int_i \tau_i G_i\left(p; \int_i \frac{\tau_i}{\sigma_i^2}\,di\right)\,di.$$

There exists a unique generalized linear equilibrium, and the price function is given by

$$P(L^*(t,z)) = G^{-1}\left[\left(1 + \frac{1}{\sigma_Z^2}\left(\int_i \frac{\tau_i}{\sigma_i^2}\,di\right)\left(\int_i \tau_i\,di\right)\right) L^*(s,z) - \overline{z}\right].$$

Proof of Proposition 2.1. Let $L(v,z) \equiv av - z$. The joint distribution of $(\widetilde{V}, \widetilde{S}_i, L(\widetilde{V}, \widetilde{Z}))$ is

$$\begin{aligned}
dF_{\widetilde{V},\widetilde{S}_i,\widetilde{L}}(v,s_i,\ell) &= dF_{\widetilde{V}}(v) f_{\widetilde{S}_i,\widetilde{L}|\widetilde{V}}(s_i,\ell|v)\,ds_i\,d\ell \\
&= dF_{\widetilde{V}}(v) f_{\widetilde{S}_i|\widetilde{V}}(s_i|v)\,ds_i\; f_{\widetilde{L}|\widetilde{V}}(\ell|v)\,d\ell \\
&= dF_{\widetilde{V}}(v)\frac{1}{\sqrt{2\pi\sigma_i^2}}\exp\left\{-\frac{1}{2\sigma_i^2}(s_i - v)^2\right\}\,ds_i\;\frac{1}{\sqrt{2\pi\sigma_Z^2}}\exp\left\{-\frac{1}{2\sigma_Z^2}(\ell - av)^2\right\}d\ell,
\end{aligned}$$

where the second equality uses the fact that $\widetilde{S}_i$ and $\widetilde{L}$ are conditionally independent given $\widetilde{V} = v$, and the last line substitutes in the expressions for the conditional densities, which are normal.

To determine the conditional density of $\widetilde{V}$ given $\widetilde{S}_i = s_i$ and $L(\widetilde{V}, \widetilde{Z}) = \ell$, I will use the fact that $dF_{\widetilde{V}|\widetilde{S}_i,\widetilde{L}}$ must be proportional to the joint distribution $dF_{\widetilde{V},\widetilde{S}_i,\widetilde{L}}$ derived above, where the constant of proportionality will depend on s_i and ℓ. Using the expression for the joint distribution, one has

$$\begin{aligned}
dF_{\widetilde{V}|\widetilde{S}_i,\widetilde{L}}(v|s_i,\ell) &\propto dF_{\widetilde{V},\widetilde{S}_i,\widetilde{L}}(v,s_i,\ell) \\
&\propto dF_{\widetilde{V}}(v)\exp\left\{-\frac{1}{2\sigma_i^2}(s_i - v)^2\right\}\,ds_i\;\exp\left\{-\frac{1}{2\sigma_Z^2}(\ell - av)^2\right\}d\ell \\
&\propto dF_{\widetilde{V}}(v)\exp\left\{\left(\frac{1}{\sigma_i^2}s_i + \frac{a}{\sigma_Z^2}\ell\right)v - \frac{1}{2}\left(\frac{1}{\sigma_i^2} + \frac{a^2}{\sigma_Z^2}\right)v^2\right\}\,ds_i\,d\ell.
\end{aligned}$$

The last line divides terms out of the exponentials that depend only on s_i or ℓ and collapses into a single exponential.

Integrating this expression gives the constant of proportionality

$$\left[\int_{\mathcal{V}} \exp\left\{\left(\frac{1}{\sigma_i^2}s_i + \frac{a}{\sigma_Z^2}\ell\right)x - \frac{1}{2}\left(\frac{1}{\sigma_i^2} + \frac{a^2}{\sigma_Z^2}\right)x^2\right\}dF_{\widetilde{V}}(x)\right]^{-1}.$$

Putting the previous two steps together, the conditional distribution of $\widetilde{V}$ is

$$dF_{\widetilde{V}|\widetilde{S}_i,\widetilde{L}}(v|s_i,\ell) = \frac{\exp\left\{\left(\frac{1}{\sigma_i^2}s_i + \frac{a}{\sigma_Z^2}\ell\right)v - \frac{1}{2}\left(\frac{1}{\sigma_i^2}+\frac{a^2}{\sigma_Z^2}\right)v^2\right\}dF_{\widetilde{V}}(v)}{\int_{\mathcal{V}}\exp\left\{\left(\frac{1}{\sigma_i^2}s_i + \frac{a}{\sigma_Z^2}\ell\right)x - \frac{1}{2}\left(\frac{1}{\sigma_i^2}+\frac{a^2}{\sigma_Z^2}\right)x^2\right\}dF_{\widetilde{V}}(x)}. \tag{3}$$

This distribution is of the exponential family form with the functions appearing in $\hat{L}(s_i,\ell) = k_{i1}(a)s_i + k_{i2}(a)b_i(\ell;a)$ given by

$$\begin{aligned} k_{i1}(a) &= \frac{1}{\sigma_i^2} \\ k_{i2}(a) &= \frac{a}{\sigma_Z^2} \\ b_i(\ell,a) &= \ell, \end{aligned}$$

and with

$$\begin{aligned} g_i(\hat{L};a) &= \log\left(\int_{\mathcal{V}}\exp\left\{\hat{L}x - \frac{1}{2}\left(\frac{1}{\sigma_i^2}+\frac{a^2}{\sigma_Z^2}\right)x^2\right\}dF_{\widetilde{V}}(x)\right) \\ dH_i(v;a) &= \exp\left\{-\frac{1}{2}\left(\frac{1}{\sigma_i^2}+\frac{a^2}{\sigma_Z^2}\right)v^2\right\}dF_{\widetilde{V}}(v), \end{aligned}$$

or, equivalently,

$$H_i(v;a) = \int_{-\infty}^{v}\exp\left\{-\frac{1}{2}\left(\frac{1}{\sigma_i^2}+\frac{a^2}{\sigma_Z^2}\right)x^2\right\}dF_{\widetilde{V}}(x).$$

The price reaction functions $G_i(p;a)$ are not available in closed form without further assumptions on the distribution $F_{\widetilde{V}}$ but are characterized as the inverse of the derivative of g_i

$$g_i'(\hat{L};a) = \frac{\int_{\mathcal{V}} x\exp\left\{\hat{L}\,x - \frac{1}{2}\left(\frac{1}{\sigma_i^2}+\frac{a^2}{\sigma_Z^2}\right)x^2\right\}dF_{\widetilde{V}}(x)}{\int_{\mathcal{V}}\exp\left\{\hat{L}\,x - \frac{1}{2}\left(\frac{1}{\sigma_i^2}+\frac{a^2}{\sigma_Z^2}\right)x^2\right\}dF_{\widetilde{V}}(x)}.$$

Given the exponential family beliefs, as long as $p \in (\underline{V},\overline{V})$ and the conditional mgf of $\widetilde{V}$ converges on an open set, Lemma A.4 implies that the demand function of each investor i is characterized by a FOC. Owing to the presence of the $\exp\{-\frac{1}{2}\left(\frac{1}{\sigma_i^2}+\frac{a^2}{\sigma_Z^2}\right)v^2\}$ term in the distribution in eq. (3), the mgf in fact converges for any $u \in \mathbb{R}$. Hence, after computing expected utility and writing down the FOC, the demand function is

$$\begin{aligned} X_i(s_i,\ell,p) &= \tau_i\left(k_{i1}(a)s_i + k_{i2}(a)\ell - G_i(p;a)\right) \\ &= \tau_i\left(\frac{1}{\sigma_i^2}s_i + \frac{a}{\sigma_Z^2}\ell - G_i(p;a)\right). \end{aligned}$$

Impose the market clearing condition by integrating across all investors and equating to the supply

$$\begin{aligned}
&\int_i \tau_i \left(\frac{1}{\sigma_i^2} s_i + \frac{a}{\sigma_Z^2} \ell - G_i(P;a) \right) di = \overline{z} + z \\
\Rightarrow &\left(\int_i \frac{\tau_i}{\sigma_i^2} di \right) v + \ell \int_i \tau_i \frac{a}{\sigma_Z^2} di - \int_i \tau_i G_i(P;a) di = \overline{z} + z \\
\Rightarrow &P = G^{-1} \left(\left(\int_i \frac{\tau_i}{\sigma_i^2} di \right) v - z + \ell \int_i \tau_i \frac{a}{\sigma_Z^2} di - \overline{z}; a \right)
\end{aligned}$$

where the second line imposes the law of large numbers to integrate over the s_i and the third line rearranges and inverts the aggregate price reaction function $G(p;a) = \int_i \tau_i G_i(p;a)$ to isolate P.

In order for this price to arise from a generalized linear price function – that is, to depend only on ℓ – it is required that $\ell = \left(\int_i \frac{\tau_i}{\sigma_i^2} di \right) v - z$. Thus, one can conclude that the equilibrium value a^* of the coefficient in the linear statistic L is uniquely defined as

$$a^* = \int_i \frac{\tau_i}{\sigma_i^2} di.$$

With a^* pinned down, the price can be simplified

$$P = G^{-1} \left[\left(1 + \frac{1}{\sigma_Z^2} \left(\int_i \frac{\tau_i}{\sigma_i^2} di \right) \left(\int_i \tau_i di \right) \right) \ell - \overline{z}; a^* \right]$$

Defining $G(p) = G(p;a^*)$ produces the expression in the Proposition.

To demonstrate that such a price function in fact exists, it suffices to show that $G^{-1}(\cdot;a^*)$ is defined on the entire real line so that regardless of the realization ℓ, the above P exists. Note that because investor beliefs are of the exponential family form, an argument identical to that used in Lemma A.5 implies that each g_i' is a monotone function that maps the entire real line onto the set $(\underline{V}, \overline{V})$. Hence, the inverses $G_i = (g_i')^{-1}$ are monotone maps of $(\underline{V}, \overline{V})$ onto $\mathbb{R}$ so that $G = \int_i \tau_i (g_i')^{-1} di$ is also a monotone map from $(\underline{V}, \overline{V})$ onto $\mathbb{R}$. Inverting once again, G^{-1} is monotone and maps $\mathbb{R}$ onto $(\underline{V}, \overline{V})$. Hence, G^{-1} is monotone, is well-defined on the entire real line, and the resulting price lies in $(\underline{V}, \overline{V})$, as required. □

In this case the coefficient in the linear statistic L is available in closed form and is unique. Therefore, there is a unique equilibrium within the generalized linear class. However, an even stronger result can be shown. The following Proposition demonstrates uniqueness of the above equilibrium among continuous equilibria, under an additional restriction on the partial derivatives of P with respect to z. Its derivation is analogous to that of Lemma 3.7 and Proposition 3.8 in the two types case. The key property of the continuum economy with normally-distributed errors

is that regardless of the functional form of the (continuous) price each investor's demand function is linear in her private signal. In particular, given any continuous price function $P(v, z)$, each investor's first-order condition reduces to

$$\int_{\mathcal{V}} (v - p) \exp\left\{ -\left(\frac{1}{\tau_i} X_i(s_i, p) - \frac{s_i}{\sigma_i^2} \right) v \right\} f_{P(\cdot)|\widetilde{V}}(p|v) \exp\left\{ -\frac{1}{2\sigma_i^2} v^2 \right\} f_{\widetilde{V}}(v)\, dv = 0,$$

after dividing out terms that are constant with respect to v. This implies that $\frac{1}{\tau_i} X_i(s_i, p) - \frac{s_i}{\sigma_i^2}$ can be written purely as a function of p. Letting $h_{i,P(\cdot)}(p)$ denote this function, which depends on the price function under consideration, and rearranging yields $X_i(s_i, p) = \tau_i \left(\frac{s_i}{\sigma_i^2} + h_{i,P(\cdot)}(p) \right)$ which is linear in the signal. The following Proposition establishes this and records the uniqueness result.

Proposition 2.2. *Suppose that Assumption 1 holds in the continuum economy with an additive, conditionally Gaussian signal structure. Suppose that $\widetilde{V}$ and $\widetilde{Z}$ are continuously distributed on intervals and have density functions. Let P be a continuous price function and suppose that for all $v \in \mathcal{V}$ one has $\frac{\partial P}{\partial z} \neq 0$ for almost every $z \in \mathcal{Z}$.*[5] *Then P must be of the generalized linear form.*

Proof of Proposition 2.2. The proof proceeds similarly to that of Proposition 3.8. Let $P(v, z)$ be any continuous equilibrium price function that depends only on the realizations of the aggregate variables v and z and satisfies the condition on the partial derivative with respect to z given in the statement of the Proposition. I will first show that this price function must reveal *at least* a linear statistic of $\widetilde{V}$ and $\widetilde{Z}$.

To derive the conditional distribution of an arbitrary investor, one must characterize the joint distribution of $(\widetilde{V}, \widetilde{S}_i, P(\widetilde{V}, \widetilde{Z}))$. I begin by demonstrating that the conditional distribution of P given $\widetilde{V} = v$ is absolutely continuous and therefore has a density. First, note that because any continuous equilibrium must be strictly monotone in z for each fixed v (see Lemma A.11 in the main text), the assumption that for all v, $\frac{\partial P}{\partial z} \neq 0$ a.e. implies that the inverse $P^{-1}(p|v)$ is locally absolutely continuous in p for each fixed v (see, e.g., (Bogachev, 2000, Exercise 5.8.54 (ii))). Furthermore, since $\widetilde{Z}$ has a density, its cdf F_Z is absolutely continuous. The conditional distribution

[5] The condition on the derivative of P ensures that the conditional distribution of P given $\widetilde{V}$ has a density function. It rules out price functions that are singularly continuous in the z variable (Cantor-like functions). As such price functions seem economically implausible and because giving the matter a full treatment would substantially complicate the proof I do not pursue it here. I conjecture that the Proposition remains true without this restriction. Note also that this condition does not presuppose that P is continuously differentiable. An argument similar to that used for Lemma A.11 in the main text implies that any continuous $P(v, z)$ is strictly monotone in z for each fixed v. Hence, for each fixed v, $\frac{\partial P}{\partial z}$ exists for almost every z and is weakly positive when it does (Royden, 1988, Ch. 5, Thm. 3).

of P given $\widetilde{V} = v$ has a cdf

$$\begin{aligned}\mathbb{P}(P(\widetilde{V}, \widetilde{Z}) \leq p | \widetilde{V} = v) &= \mathbb{P}(\widetilde{Z} \geq P^{-1}(p|v)) \\ &= 1 - F_{\widetilde{Z}}(P^{-1}(p|v)).\end{aligned}$$

This cdf is a composition of a bounded absolutely continuous function $F_{\widetilde{Z}}$ and a monotone locally absolutely continuous function, and it is straightforward to show that this implies that it is itself absolutely continuous on its domain $\{P(z, v) : z \in \mathcal{Z}\}$.

The joint distribution of $(\widetilde{V}, \widetilde{S}_i, P(\cdot))$ thus satisfies

$$\begin{aligned}dF_{\widetilde{V},\widetilde{S}_i,P(\cdot)}(v, s_i, p) &= dF_{\widetilde{S}_i,P(\cdot)|\widetilde{V}}(s_i, p|v) f_{\widetilde{V}}(v)\, dv \\ &= dF_{\widetilde{S}_i|\widetilde{V}}(s_i|v) dF_{P(\cdot)|\widetilde{V}}(p|v) dF_{\widetilde{V}}(v) \\ &= \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left\{-\frac{1}{2\sigma_i^2}(s_i - v)^2\right\} ds_i \; f_{P(\cdot)|\widetilde{V}}(p|v)\, dp \; f_{\widetilde{V}}(v)\, dv, \qquad (4)\end{aligned}$$

where the second equality uses the fact that the conditional independence of $\widetilde{\varepsilon}_i$ and $\widetilde{Z}$ implies conditional independence of $\widetilde{S}_i$ and $P(\widetilde{V}, \widetilde{Z})$ and the final equality uses the fact that all distributions in the previous line have density functions and substitutes in explicitly for the conditional density of the signal.

Note that Lemma A.4 implies that for $p \in (\underline{V}, \overline{V})$ (which are the only prices that are consistent with equilibrium) investors' demand functions are characterized by a first-order condition owing to the fact that the conditional mgf

$$M_{\widetilde{V}|\widetilde{S}_i,P(\cdot)}(u|s_i, p) = \int_{\mathcal{V}} \exp\{uv\} \frac{\frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left\{-\frac{1}{2\sigma_i^2}(s_i - v)^2\right\} f_{P(\cdot)|\widetilde{V}}(p|v) f_{\widetilde{V}}(v)}{\int_{\mathcal{V}} \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left\{-\frac{1}{2\sigma_i^2}(s_i - x)^2\right\} f_{P(\cdot)|\widetilde{V}}(p|x) f_{\widetilde{V}}(x)\, dx} dv \qquad (5)$$

converges on the entire real line owing to the presence of the $\exp\left\{-\frac{1}{2\sigma_i^2}v^2\right\}$ term in the integral in the numerator.

Hence, to characterize investor demand, it is sufficient to find $X_i(s_i, p)$ that satisfies the FOC

$$\int_{\mathcal{V}} (v - p) \exp\left\{-\frac{1}{\tau_i} X_i(s_i, p)(v - p)\right\} f_{\widetilde{V}|\widetilde{S}_i,P(\cdot)}(v|s_i, p) = 0.$$

By dividing out terms that are constant with respect to v the following condition is equivalent

$$\int_{\mathcal{V}} (v - p) \exp\left\{-\frac{1}{\tau_i} X_i(s_i, p) v\right\} f_{\widetilde{V},\widetilde{S}_i,P(\cdot)}(v, s_i, p) = 0, \qquad (6)$$

which uses the fact that the joint density is proportional to the conditional, where the constant of proportionality depends on s_i and p but not v.

Plug the joint distribution from eq. (4) into eq. (6) and again divide out terms that are constant with respect to v

$$\int_{\mathcal{V}} (v-p) \exp\left\{ -\left(\frac{1}{\tau_i} X_i(s_i,p) - \frac{s_i}{\sigma_i^2} \right) v \right\} f_{P(\cdot)|\widetilde{V}}(p|v) \exp\left\{ -\frac{1}{2\sigma_i^2} v^2 \right\} f_{\widetilde{V}}(v)\, dv = 0.$$

Let $\check{X}_i(s_i,p) = \frac{1}{\tau_i} X_i(s_i,p) - \frac{s_i}{\sigma_i^2}$. With this substitution, the FOC implies that $\check{X}_i(s_i,p)$ can be written purely as a function of p as no s_i appears elsewhere in the expression:

$$\int_{\mathcal{V}} (v-p) \exp\left\{ -\check{X}_i(s_i,p) v \right\} dF_{P(\cdot)|\widetilde{V}}(p|v) \exp\left\{ -\frac{1}{2\sigma_i^2} v^2 \right\} f_{\widetilde{V}}(v)\, dv = 0.$$

Let $h_{i,P(\cdot)}(p)$ denote this function of p and rearrange the definition of $\check{X}_i$ to obtain

$$X_i(s_i,p) = \tau_i \left(\frac{s_i}{\sigma_i^2} + h_{i,P(\cdot)}(p) \right).$$

Thus, regardless of the functional form of the price, each investor's demand function is linear in her signal. The market-clearing condition therefore implies that *any* price function must reveal to investors at least the value of the statistic

$$L = \int_i \tau_i \frac{\widetilde{S}_i}{\sigma_i^2} - \widetilde{Z} = \widetilde{V} \left[\int_i \frac{\tau_i}{\sigma_i^2} \right] - \widetilde{Z},$$

where I use the law of large numbers to cancel the idiosyncratic errors.

Now, focusing on continuous price function, an identical argument to that applied to derive Lemma 3.7 shows that any continuous price function must depend *only* on the statistic L and must therefore be a generalized linear price function. To save space, I do not repeat the argument here. □

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